

MADELYN MATHAI

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SUMMARY

PhD student in Data Science at the University of Virginia whose core research is technical AI explainability, with a focus on mechanistic interpretability for AI safety. My work examines how the internal representations of AI models can be verified, causally tested, and selectively removed, across models of different modalities. I have also applied these methods to AI models of climate systems, helping introduce them to the climate science community. Beyond my core research, I apply data science to civic and equity-focused work with nonprofits, and I aim to engage more with AI policy, using technical research to inform it.

EDUCATION

University of Virginia, School of Data Science
Doctor of Philosophy in Data Science

Expected May 2029

University of Pennsylvania, College of Arts and Sciences
Bachelor of Arts

2018 – 2020

Major: Economics; Minors: Mathematics, Survey Research & Data Analytics
Honors: Cum Laude

PUBLICATIONS

Mathai, M., Higgins, T. B., Grise, K. M., Agarwal, C., & Mamalakis, A. (2026). *Mechanistic Interpretability of Atmospheric Rivers in GraphCast* [Spotlight Paper]. NeurIPS 2026 Tackling Climate Change with Machine Learning Workshop.

Ghosh, S., **Mathai, M.**, Radhakrishnan, V. B., & Agarwal, C. (2026). *Diminishing Returns of Data in Sparse Autoencoders: Functional Behavior does not Guarantee Feature Reproducibility* [Manuscript submitted for publication].

TEACHING EXPERIENCE

Graduate Teaching Assistant, University of Virginia: Introduction to Data Science (Fall 2024), Introduction to Programming (Spring 2025)

RESEARCH EXPERIENCE

Data Efficiency in Sparse Autoencoders

June 2026 – Aug 2026

Under supervision of Dr. Chirag Agarwal (Aikyam Lab)

- Trained sparse autoencoders across 3 language models, 4 architectures, and budgets from 5M to 500M tokens, showing reconstruction nears full-data performance at 5% of the budget
- Found that feature reproducibility keeps improving after functional metrics plateau, with strict one-to-one feature matches rising from 2% at 1% of the data to 29% at 10%
- Compared data selection strategies, finding that diversity-based sampling improves automated interpretability and unlearning while random sampling preserves reconstruction

Mechanistic Interpretability of Atmospheric Rivers in GraphCast

Jan 2026 – Aug 2026

Under supervision of Dr. Chirag Agarwal (Aikyam Lab) and Dr. Antonios Mamalakis (Mamalakis Lab)

- Applied sparse autoencoders to GraphCast, a state-of-the-art AI weather model, surfacing the physical features driving its predictions
- Introduced matryoshka sparse autoencoders to the climate science community as a tool for learning hierarchical, multi-scale feature dictionaries
- Traced atmospheric river concepts across depth and demonstrated causality by feature steering

FELLOWSHIPS

Community Data Fellowship, University of Virginia

2025 – Present

One of four graduate fellows selected to lead a community-centered data science project with the Center for Community Partnerships. Partnering with The Fountain Fund, a nonprofit providing low-interest loans to formerly incarcerated people, to deliver data analysis supporting their lending and advocacy work. Award: \$7,000.

Fox Research and Service Fellowship, University of Pennsylvania

2019 – 2020

Awarded by Robert A. Fox Leadership Program

Selected to work with Penn Opinion Research Election Studies faculty on research analyzing U.S. political outcomes using public opinion and election data. Award: \$2,875.

EMPLOYMENT

Ubisoft

2021 – 2024

Junior Game Data Analyst

- Conducted in-depth analysis of matchmaking rating with Python and presented to project leadership
- Created 50+ ETL pipelines with HiveQL that fed into Tableau dashboards or ad-hoc presentations for executive leadership
- Developed a singular value decomposition-based recommender system for in-game purchases

Cambridge Associates

2020 – 2021

Investment Analyst

- Conducted due diligence on managers and presented recommendations at client meetings
- Wrote Python scripts to automate Excel processes to create client-ready data
- Produced data visualizations on investment performance and asset allocation for 30+ investment committee meetings

SERVICE

Standing Committee on AI in Teaching and Learning, UVA Provost's Office

2026 – Present

Only student on a university-wide committee advising on AI policy, pedagogy, and responsible use in education.

Volunteer Data Associate, Correctional Association of New York

2023

Created a text extraction tool with Python to pull text from images of death records within New York prisons.